

Week 48 Inside and outside covers

Facilitator guide | Final student packets v2 | Prepared 3 October 2026 | Unpiloted

Mathematical overview

Finite area certificates. The sum of grid-square areas wholly inside a shape is a lower bound; the sum for a grid-square cover is an upper bound. Use aligned tiles without overlap of positive area. For the polygonal shapes here, count a cell in the cover only when it contains positive gray area. Edge-only and corner-only contact adds no cell; the boundary is already covered by neighboring closed tiles.

On nested refinement, a whole inside cell keeps all its area, while a cover cell can keep all four children or discard the outside ones. Therefore the lower bound cannot fall and the upper bound cannot rise. Their gap is an explicit uncertainty certificate. Equality of limiting inner and outer contents requires a measurability condition; finite refinement alone does not prove it for every set.

Keep the unit fixed. One big square has area 1; four half-side squares have total area 1. The main mats are 120 mm square, with 30 mm big tiles and 15 mm small tiles. A small tile has area $1/4$, not $1/2$. All numerical bounds in the older packets use big-square units.

Discovery by band. K-1 classifies cells, selects useful refinements and changes area without changing a coarse certificate. Grades 2-3 tightens bounds and defeats claims of exact area. Grades 4-5 optimizes a limited refinement budget and explains monotonicity. Counting suggests a pattern; cell containment proves the bounds.

Prepare and launch

Per table: chosen student pages, pencils or two colors, transparent sleeves or tracing paper, 16 paper 30 mm squares and at least 8 paper 15 mm squares. For complete fine covers, shade rather than prepare 40 loose tiles. Use paper, not an assumed commercial tile size. Print at actual size; the small summary triangles on the last upper page are explicitly not to scale.

Readiness. Start with visual whole/partial/outside decisions and adult counting. Four-small-to-one-big regrouping is enough for older bounds; fraction symbols are optional. Keep one adult at each of the current three tables.

Launch, 4 minutes. Let children cover a few cells. Use the three printed sample cells to show whole gray, partly gray and outside. Demonstrate an edge touch that needs no extra tile, then reassemble four small squares into one big one. Ask, "What can your tiles prove about the amount of gray?"

Flexible hour. 0-5 launch; 5-22 coarse covers; 22-25 stand and sort large imaginary cells; 25-43 refine or change a shape; 43-55 compare certificates and counterexamples; 55-60 reset. One sound before-and-after cover is enough for a useful session.

Shared triangle and diamond keys

Final task	Whole inside cells	Cover cells	Bounds in big units
All bands P1 triangle	6 big	10 big	$6 \leq \text{area} \leq 10$
Older P2 diamond	4 big	12 big	$4 \leq \text{area} \leq 12$
All bands P3 triangle	28 small	36 small	$7 \leq \text{area} \leq 9$
Older P4 diamond	24 small	40 small	$6 \leq \text{area} \leq 10$

Problem 1. For the coarse triangle, successive rows from top to bottom contain 0,1,2,3 whole cells and 1,2,3,4 cover cells. Four diagonal cells are partial. **Older Problem 2:** the diamond has four central whole cells and eight partial cells. Therefore its area is certainly more than 3 and less than 13; an exact-area formula is unnecessary.

Problem 3. Each of the four partial big triangle cells becomes one whole, two partial and one outside small cell. This adds one big unit to the inside bound and removes one from the cover.

Older Problem 4: each of the eight partial big diamond cells has the same four-child pattern. Its uncertainty gap falls from 8 to 4 big units. The triangle's gap falls from 4 to 2.

K-1 specific keys

Problem 2. The trapezoid's coarse cover uses all 16 big cells. Refine **any two of the four corner cells**: each has one partial child and three outside children, so remove **6 small tiles** in total. No covered big cell can lose all four children, so three removable children per chosen tile is the absolute maximum. The remaining cover has area $16 - 6/4 = 14.5$ big units; children may simply count the removed pieces.

Problem 4. Each of the diamond's eight partial big cells has one wholly outside small child. Choose four such children in different big cells and bulge the nearby diagonal boundary slightly into each, leaving part of each chosen small cell unshaded. Keep all changes within those four partial big cells. Exactly four previously outside small cells can become partial, while every big-cell status stays unchanged. There is no unique changed shape.

Problem 5. In the same three columns, shade a rectangle one full row high plus a shallow part of the next row. Width 3 and height $7/6$ gives area **3.5**; width 3 and height $11/6$ gives **5.5**. Both have exactly three whole and the same three partial big cells. Folding or drawing a visibly thin versus nearly full extra row suffices; children need not measure sixths.

Problem 6. Yes. Raise the $11/6$ -height rectangle to height $23/12$, for example: area 5.75. Its three partial cells remain partial. One can approach area 6 as closely as desired while leaving a positive unshaded strip, but cannot attain 6 while all three remain genuinely partial. Thus 6 is a **supremum, not a largest allowed area**.

Older specific keys

Grades 2-3 Problem 5. Width-4 rectangles with heights $9/8$ and $15/8$ have areas **4.5 and 7.5**. Both use one whole row and part of the next: exactly four whole and four partial big cells, in the same positions.

Grades 2-3 Problem 6. Neither less than 4 nor more than 8 is possible. The coarse certificate is $4 \leq \text{area} \leq 8$. For ordinary drawings with positive gray and positive white area in each partial cell, actually $4 < \text{area} < 8$. These are strict because some gray must be added beyond the four whole cells and some white must remain below the eight-cell cover. Areas can approach either endpoint; neither is attained with four genuinely partial cells.

Grades 4-5 Problem 5. Number columns 1-4 from left to right and rows 1-4 from top to bottom. Refine any three of **(2,1), (3,1), (2,4), (3,4)**. Each of these top/bottom middle cells splits into two whole and two outside small cells, eliminating its entire one-unit uncertainty. The initial bounds 4 and 16 become **5.5 and 14.5**, so the gap falls from 12 to **9**.

Trapezoid cell type	Small children I / P / O	Gap reduction per big cell
Top/bottom middle, 4 cells	2 / 0 / 2	1
Four corner cells	0 / 1 / 3	3/4
Left/right middle, 4 cells	0 / 2 / 2	1/2
Central whole, 4 cells	4 / 0 / 0	0

Here I/P/O means inside/partial/outside. A big cell can remove at most its entire one-unit gap, so three refinements cannot gain more than 3. The displayed choice achieves 3 and proves optimality. Notice that this objective differs from K-1's removal-only objective, where corners are best.

Grades 4-5 Problem 6. No to both questions. Every old inside square is exactly its four inside children. New inside children may add area, never remove old inside area. Every retained cover child lies in its parent cover square. Deleting outside children can only reduce the cover. This assumes the same shape and a true subdivision of the same grid.

Grades 4-5 Problem 7. Two congruent triangles fill the 4-by-4 square, whose area is 16. The triangle's exact area is **8 big-square units**, lying between 6 and 10 and between 7 and 9. Refinement improves the certificate, not the shape's actual area.

Optional hints and sources

If stuck, ask "Which uncertain tile would tell us the most if split?" Keep inside and cover colors separate. For shape-changing tasks, preserve the same labeled cells and try thin strips. These are optional hints, not a mandatory method.

A. Sapozhnikov, *Mathematics 3*, §1.2, pp. 2-4, develops inner/outer Jordan content and nested dyadic boxes: [author lecture notes](#). Our finite boundary-touch convention and all polygon counts, optimal refinements and rectangle witnesses were checked independently by exact rational clipping. All material handling and pacing are unpiloted proposals; no physical testing is claimed.